

Regression

Truth is much too complicated to allow anything but approximations.

—JOHN VON NEUMANN

In Chapter 2, we considered the conditional expectation function (CEF) and best linear predictor (BLP) as ways to summarize bivariate or multivariate relationships. In this chapter, we apply the principles on estimation from Chapter 3 to the CEF and BLP in order to derive a simple and powerful tool for empirical practice: regression. In its simplest form, regression can be seen as a method for estimating the BLP using the data in our sample, much as the sample mean is an estimator for the population mean. More generally, we see that regression estimators provide a flexible way to approximate the CEF in a principled and transparent manner.

The regression estimator is central in both theoretical and applied statistics and econometrics, in part because it can be motivated by many different assumptions and statistical perspectives. For example, regression can be motivated by a restrictive parametric model known as the classical linear model, which we will discuss in Sections 5.1.1 (*The Classical Linear Model*) and 5.2.1 (*The Logic of Maximum Likelihood Estimation*). Motivations for regression abound—including motivations that even require no probability theory at all (see, e.g., Hastie, Tibshirani, and Friedman 2009, Ch. 1). Our treatment of regression focuses on our agnostic approach: although we assume i.i.d. sampling (and the usual plug-in regularity conditions), we require no assumptions about the functional form of the CEF.

4.1 REGRESSION ESTIMATION

We begin by showing how regression is a simple plug-in estimator for the BLP. We know from Section 3.3 (*The Plug-In Principle*) that plug-in estimators provide a rather general way to estimate population functionals, and the regression estimator is no exception.

4.1.1 Bivariate Case

Let us first consider the bivariate case, where we are interested in the random vector (X, Y) . Recall that the BLP of Y given X is $g(X) = \alpha + \beta X$, where

$$\alpha = E[Y] - \frac{\text{Cov}[X, Y]}{V[X]}E[X],$$

$$\beta = \frac{\text{Cov}[X, Y]}{V[X]}.$$

Applying Theorems 2.1.13 (*Alternative Formula for Variance*) and 2.2.2 (*Alternative Formula for Covariance*), these equations can be rewritten in terms of expected values:

$$\alpha = E[Y] - \frac{E[XY] - E[X]E[Y]}{E[X^2] - E[X]^2}E[X],$$

$$\beta = \frac{E[XY] - E[X]E[Y]}{E[X^2] - E[X]^2}.$$

We can then *estimate* the BLP using plug-in estimation:

$$\hat{\alpha} = \bar{Y} - \frac{\overline{XY} - \bar{X} \cdot \bar{Y}}{\overline{X^2} - \bar{X}^2} \bar{X},$$

$$\hat{\beta} = \frac{\overline{XY} - \bar{X} \cdot \bar{Y}}{\overline{X^2} - \bar{X}^2}.$$

Thus, we can use *sample* data to estimate the population relationship: $\hat{\alpha}$ is the regression estimate of the intercept of the BLP and is consistent for α , and $\hat{\beta}$ is the regression estimate of the slope of the BLP and is consistent for β . (Consistency follows immediately from the Weak Law of Large Numbers and Continuous Mapping Theorem.)

Example 4.1.1. *Estimating the BLP*

Suppose we have a random vector (X, Y) such that

$$Y = 1 + 2X + 5U,$$

where X and U are independent and each distributed as $N(0, 1)$. We can derive the CEF of Y given X :

$$E[Y|X] = E[1 + 2X + 5U|X] = 1 + 2E[X|X] + 5E[U] = 1 + 2X.$$

In this example, the CEF is linear, so the BLP *is* the CEF, and so we have $\alpha = 1$, $\beta = 2$. Now, suppose we do not know the joint distribution of (X, Y) and we wanted to estimate the CEF of Y given X from sample data. We would observe a sample of draws from (X, Y) ; such a sample is shown in Table 4.1.1.

TABLE 4.1.1. Sample of $n = 15$ Draws from (X, Y)

Unit	Y	X
1	6.88	1.37
2	1.17	0.30
3	16.16	1.57
4	-4.84	-0.69
5	-6.17	-0.64
6	0.67	0.08
7	3.36	0.47
8	-0.69	1.33
9	8.03	-0.13
10	-3.35	0.59
11	1.87	-0.44
12	13.54	0.76
13	4.06	0.73
14	6.72	-3.54
15	7.20	-0.05

The dataset shown in the table is all that is available to the researcher. The goal is to estimate the CEF, but the CEF is hard to characterize—it could have *any* functional form.¹ From our fifteen data points, it is impossible to distinguish all the different shapes that the CEF could take. (We will provide further discussion of this point in Section 4.3.2.) Nevertheless, regression consistently estimates the minimum mean squared error (MSE) *linear* approximation of the CEF, that is, the BLP.

In Figure 4.1.2, we overlay the data points, the actual (unknown to the researcher) CEF (in black) and the regression estimate of the BLP (in red). We see that, even with an n of 15, we approximate the relationship fairly well. Naturally, if we were to add more data, we would expect that this approximation would improve accordingly. $\triangle$

4.1.2 OLS Regression

Suppose now that we have K explanatory variables: $X_{[1]}, X_{[2]}, \dots, X_{[K]}$. In this case, as we saw in Section 2.3 (*Multivariate Generalizations*), the BLP is still defined as the linear function that minimizes MSE. That is, the BLP (also known as the *population regression function*) is the linear function

¹ In this case, we know that it is in fact linear, but we are pretending we do not know that.

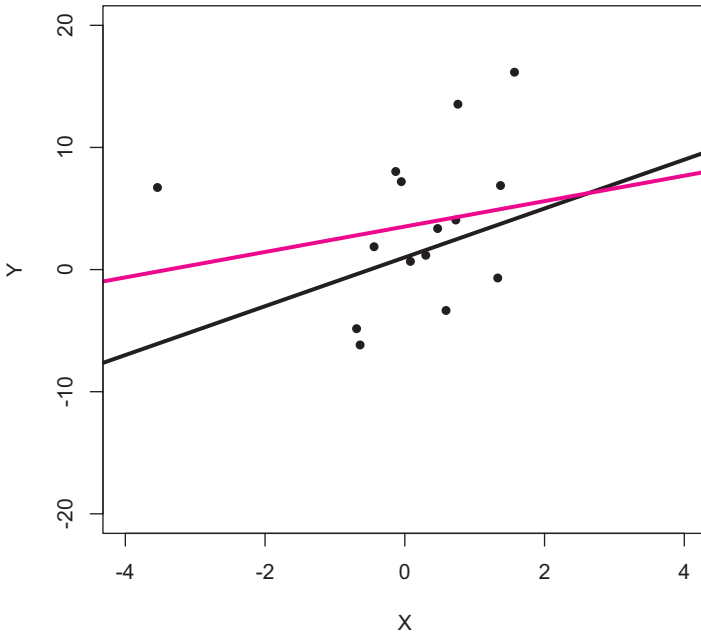


FIGURE 4.1.2. Illustration of $n = 15$ Sample of Draws from (X, Y)

$g(X_{[1]}, X_{[2]}, \dots, X_{[K]}) = \beta_0 + \beta_1 X_{[1]} + \beta_2 X_{[2]} + \dots + \beta_K X_{[K]}$ such that

$$(\beta_0, \beta_1, \dots, \beta_K) =$$

$$\underset{(b_0, b_1, \dots, b_K) \in \mathbb{R}^{K+1}}{\operatorname{argmin}} \mathbb{E} \left[\left(Y - (b_0 + b_1 X_{[1]} + b_2 X_{[2]} + \dots + b_K X_{[K]}) \right)^2 \right].$$

How would we estimate the BLP? Unfortunately, in the multivariate case, we no longer have the simple formulas $\alpha = \mathbb{E}[Y] - \frac{\operatorname{Cov}[X, Y]}{\operatorname{V}[X]} \mathbb{E}[X]$ and $\beta = \frac{\operatorname{Cov}[X, Y]}{\operatorname{V}[X]}$. But the plug-in principle still applies; we can still replace expectations with sample means in the definition of the BLP to obtain an estimate thereof.²

Definition 4.1.2. OLS Regression Estimator

For i.i.d. random vectors $(Y_1, \mathbf{X}_1), (Y_2, \mathbf{X}_2), \dots, (Y_n, \mathbf{X}_n)$, the *ordinary least squares (OLS) regression estimator* is the function $\hat{g}(\mathbf{X}) = \hat{\beta}_0 + \hat{\beta}_1 X_{[1]} +$

² Note that the term “multivariate regression” does not refer to regression with more than one explanatory variable; rather it refers to the case where the outcome is a random vector. The multivariate generalization of regression we consider here is often referred to as “multiple regression.”

$\hat{\beta}_2 X_{[2]} + \dots + \hat{\beta}_K X_{[K]}$ such that

$$\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_K)$$

$$= \underset{(b_0, b_1, \dots, b_K) \in \mathbb{R}^{K+1}}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n \left(Y_i - (b_0 + b_1 X_{[1]i} + b_2 X_{[2]i} + \dots + b_K X_{[K]i}) \right)^2.$$

We refer to $\hat{\beta}$ as the *OLS coefficient estimator*.

The quantity $e_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_{[1]i} + \hat{\beta}_2 X_{[2]i} + \dots + \hat{\beta}_K X_{[K]i})$ is known as the i^{th} *residual*. It is the difference between the observed value of Y_i and the value we would have predicted given the observed values of $X_{[1]i}$, $X_{[2]i}$, ..., $X_{[K]i}$ and our estimate of the BLP. A residual is the sample analog of a prediction *error*, which is the difference between an observed value of Y_i and the value that would have been predicted based on the observed values of $X_{[1]i}$, $X_{[2]i}$, ..., $X_{[K]i}$ and some actual population predictor (e.g, the true CEF or BLP). This estimator is called the *ordinary least squares (OLS)* regression estimator because it finds the fit that minimizes the mean of squared residuals (or, equivalently, minimizes the sum of squared residuals).³ Thus, the regression estimator is a plug-in estimator; in order to estimate the parameters that minimize the mean squared error, we find the parameters that minimize the mean squared residual.

4.1.3 Regression with Matrix Algebra

In this section, we show how to derive a closed-form representation of the OLS regression estimator using matrix algebra. We recommend that all readers (even those who are less familiar with matrix algebra) examine the proofs in this section, as they show how matrix algebra can be profitably used to simplify linear optimization problems.

Our goal is to find the coefficient values that minimize the sum of squared residuals. To do this, we will need to represent the problem in matrix notation. In particular, we will need to define the *regressor matrix* (also known as the *design matrix* or *model matrix*).

³ Strictly speaking, OLS is the oldest and most familiar regression method. The many others include weighted least squares and penalized regression (Section 4.3.6). In this book, “regression” refers to OLS unless otherwise noted.

Definition 4.1.3. Regressor Matrix

For i.i.d. random vectors $(Y_1, \mathbf{X}_1)$, $(Y_2, \mathbf{X}_2)$, ..., $(Y_n, \mathbf{X}_n)$, the *regressor matrix* for the OLS regression estimator is

$$\mathbb{X} = \begin{pmatrix} 1 & X_{[1]1} & X_{[2]1} & \cdots & X_{[K]1} \\ 1 & X_{[1]2} & X_{[2]2} & \cdots & X_{[K]2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{[1]n} & X_{[2]n} & \cdots & X_{[K]n} \end{pmatrix}.$$

(All matrices in this book shall be denoted by “blackboard bold” letters, for example, $\mathbb{X}$ and $\mathbb{W}$.)

We can now state the canonical closed-form OLS regression solution.

Theorem 4.1.4. OLS Solution in Matrix Algebra

Let $(\mathbf{X}_1, Y_1)$, $(\mathbf{X}_2, Y_2)$, ..., $(\mathbf{X}_n, Y_n)$ be i.i.d. random vectors. Then, if $\mathbb{X}^T \mathbb{X}$ is invertible, the OLS coefficient estimator is given by

$$\hat{\boldsymbol{\beta}} = (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \mathbf{Y}.$$

Proof: For any $(b_0, b_1, \dots, b_K) \in \mathbb{R}^{K+1}$ and for every observation i , we can define the i^{th} residual as

$$e_i = Y_i - (b_0 + b_1 X_{[1]i} + b_2 X_{[2]i} + \dots + b_K X_{[K]i}).$$

We can “stack” all n of these e_i s to form a system of equations,

$$e_1 = Y_1 - (b_0 + b_1 X_{[1]1} + b_2 X_{[2]1} + \dots + b_K X_{[K]1})$$

$$e_2 = Y_2 - (b_0 + b_1 X_{[1]2} + b_2 X_{[2]2} + \dots + b_K X_{[K]2})$$

$$\vdots$$

$$e_n = Y_n - (b_0 + b_1 X_{[1]n} + b_2 X_{[2]n} + \dots + b_K X_{[K]n}).$$

This system of equations can be rewritten in matrix form as:

$$\begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} - \begin{pmatrix} 1 & X_{[1]1} & X_{[2]1} & \cdots & X_{[K]1} \\ 1 & X_{[1]2} & X_{[2]2} & \cdots & X_{[K]2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{[1]n} & X_{[2]n} & \cdots & X_{[K]n} \end{pmatrix} \begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ \vdots \\ b_K \end{pmatrix},$$

or equivalently, $\mathbf{e} = \mathbf{Y} - \mathbb{X}\mathbf{b}$, where $\mathbf{e} = (e_1, e_2, \dots, e_n)^T$, $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)^T$, $\mathbf{b} = (b_0, b_1, b_2, \dots, b_K)^T$, and $\mathbb{X}$ is the regressor matrix. Thus, finding the OLS

regression estimator is equivalent to the following minimization problem:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\mathbf{b} \in \mathbb{R}^{K+1}} \sum_{i=1}^n e_i^2 = \arg \min_{\mathbf{b} \in \mathbb{R}^{K+1}} (\mathbf{e}^T \mathbf{e}) = \arg \min_{\mathbf{b} \in \mathbb{R}^{K+1}} (\mathbf{Y} - \mathbb{X}\mathbf{b})^T (\mathbf{Y} - \mathbb{X}\mathbf{b}).$$

The first-order condition (that is, setting the derivative of the sum of squared residuals with respect to the coefficients equal to zero) yields

$$-2\mathbb{X}^T(\mathbf{Y} - \mathbb{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}.^4$$

Solving for $\hat{\boldsymbol{\beta}}$,

$$\begin{aligned} -\mathbb{X}^T(\mathbf{Y} - \mathbb{X}\hat{\boldsymbol{\beta}}) &= \mathbf{0} \\ -\mathbb{X}^T\mathbf{Y} + \mathbb{X}^T\mathbb{X}\hat{\boldsymbol{\beta}} &= \mathbf{0} \\ \mathbb{X}^T\mathbb{X}\hat{\boldsymbol{\beta}} &= \mathbb{X}^T\mathbf{Y} \\ (\mathbb{X}^T\mathbb{X})^{-1}(\mathbb{X}^T\mathbb{X})\hat{\boldsymbol{\beta}} &= (\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T\mathbf{Y} \\ \hat{\boldsymbol{\beta}} &= (\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T\mathbf{Y}. \quad \square \end{aligned}$$

Note that the condition that $\mathbb{X}^T\mathbb{X}$ must be invertible implies that all the explanatory variables cannot be written as linear combinations of one another; that is, they are not *exactly multicollinear*. Such cases arise, for instance, when the number of variables $K > n$ (an extreme case of *micronumerosity*), or when some variables have no variance. In practice, this condition suffices to make the OLS solution unique. For example, in the case of a single explanatory variable, assuming invertibility of $\mathbb{X}^T\mathbb{X}$ is equivalent to assuming that $\hat{V}[X] > 0$. If this condition were not to hold, there would be infinitely many $\hat{\boldsymbol{\beta}}$ values that would all minimize the mean of squared residuals.

Some noteworthy properties of the regression estimator also follow from the matrix algebraic solution.

Theorem 4.1.5. *Properties of Residuals from the OLS Regression Estimator*

Let $(Y_1, \mathbf{X}_1), (Y_2, \mathbf{X}_2), \dots, (Y_n, \mathbf{X}_n)$ be i.i.d. random vectors. Let $\hat{\boldsymbol{\beta}}$ be the OLS coefficient estimator, and let the i^{th} residual $e_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_{[1]i} + \hat{\beta}_2 X_{[2]i} + \dots + \hat{\beta}_K X_{[K]i}), \forall i \in \{1, 2, \dots, n\}$. Then

- $\bar{e} = 0$, where $\bar{e} = \frac{1}{n} \sum_{i=1}^n e_i$.
- $\forall k \in \{1, 2, \dots, K\}, \overline{eX_{[k]}} = 0$, where $\overline{eX_{[k]}} = \frac{1}{n} \sum_{i=1}^n e_i X_{[k]i}$.
- $\forall k \in \{1, 2, \dots, K\}, \overline{eX_{[k]}} - \bar{e}\bar{X}_{[k]} = 0$.

⁴ Readers without familiarity with matrix calculus should not fret. Just take our word for it that the derivative of $(\mathbf{Y} - \mathbb{X}\hat{\boldsymbol{\beta}})^T(\mathbf{Y} - \mathbb{X}\hat{\boldsymbol{\beta}})$ with respect to $\hat{\boldsymbol{\beta}}$ is $-2\mathbb{X}^T(\mathbf{Y} - \mathbb{X}\hat{\boldsymbol{\beta}})$, noticing how this resembles the chain rule. $\mathbf{0}$ denotes the zero vector of appropriate length. We omit checking the second-order conditions.

Proof: The first-order condition $-2\mathbb{X}^T(\mathbf{Y} - \mathbb{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$ is equivalent to $\mathbb{X}^T\mathbf{e} = \mathbf{0}$, that is,

$$\begin{pmatrix} 1 & 1 & \cdots & 1 \\ X_{[1]1} & X_{[1]2} & \cdots & X_{[1]n} \\ X_{[2]1} & X_{[2]2} & \cdots & X_{[2]n} \\ \vdots & \vdots & \ddots & \vdots \\ X_{[K]1} & X_{[K]2} & \cdots & X_{[K]n} \end{pmatrix} \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

which summarizes the system of equations

$$\begin{aligned} e_1 + e_2 + \dots + e_n &= 0, \\ e_1 X_{[1]1} + e_2 X_{[1]2} + \dots + e_n X_{[1]n} &= 0, \\ e_1 X_{[2]1} + e_2 X_{[2]2} + \dots + e_n X_{[2]n} &= 0, \\ &\vdots \\ e_1 X_{[K]1} + e_2 X_{[K]2} + \dots + e_n X_{[K]n} &= 0. \end{aligned}$$

The first equation says that the sum of the residuals is zero. If the residuals did not sum to zero, we could obtain a better fit by changing the constant term. Furthermore, dividing each equation by n ,

$$\begin{aligned} \bar{e} &= \frac{1}{n} \sum_{i=1}^n e_i = 0, \\ \overline{eX_{[1]}} &= \frac{1}{n} \sum_{i=1}^n e_i X_{[1]i} = 0, \\ \overline{eX_{[2]}} &= \frac{1}{n} \sum_{i=1}^n e_i X_{[2]i} = 0, \\ &\vdots \\ \overline{eX_{[K]}} &= \frac{1}{n} \sum_{i=1}^n e_i X_{[K]i} = 0, \end{aligned}$$

which also implies that for $k = 1, 2, \dots, K$, the sample covariance between the residuals and $X_{[k]}$ is zero: $\overline{eX_{[k]}} - \bar{e}\bar{X}_{[k]} = 0$. $\square$

Naturally, these results provide sample analogs to our results on properties of deviations from the BLP as in Theorems 2.2.22 and 2.3.8. The first property, $\bar{e} = 0$, states that the residuals have mean zero, much as we know that the deviations from the BLP have expectation zero ($E[\varepsilon] = 0$). Similarly, the second property, $\overline{eX_{[k]}} = 0, \forall k \in \{1, 2, \dots, K\}$, states that the product of the

residuals and the explanatory variables also has mean zero. This is analogous to noting that in the population problem, the product of the deviations from the BLP and the explanatory variables is zero: $E[\varepsilon X_{[k]}] = 0, \forall k \in \{1, 2, \dots, K\}$. The final statement—that the (plug-in) sample covariance of the residuals and the explanatory variables is zero, or $\overline{eX_{[k]}} - \bar{e}\bar{X}_{[k]} = 0, \forall k \in \{1, 2, \dots, K\}$ —is perhaps best known, and is analogous to the statement that the covariance between the deviations from the BLP and the explanatory variables is zero ($\text{Cov}[\varepsilon, X_{[k]}] = 0, \forall k \in \{1, 2, \dots, K\}$).

4.2 INFERENCE

We have shown that we can estimate the BLP using plug-in estimation. Just like every other estimator we have seen, each of our coefficient estimators $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_K$ is a random variable. Each therefore has a sampling distribution and, thus, a sampling variance. As usual, we want to *estimate* these sampling variances so that we can construct confidence intervals and perform hypothesis tests for the values of these coefficients.

4.2.1 Standard Errors and Inference

We begin our discussion of standard errors by exploring a very simple case: suppose that we had no explanatory variables. In this setting, we seek to estimate

$$\beta_0 = \arg \min_{b_0 \in \mathbb{R}} E[(Y - b_0)^2].$$

We know from Theorem 2.1.24 (*Expected Value Minimizes MSE*) that the solution is $\beta_0 = E[Y]$. The sample analog to β_0 would then be $\bar{Y}$, so using plug-in estimation, $\hat{\beta}_0 = \bar{Y}$. Thus, the regression estimate of β_0 with no explanatory variables is the sample mean.

In this simple case, inference proceeds exactly as we saw in Chapter 3: the standard error of $\hat{\beta}_0$ is

$$\sigma[\hat{\beta}_0] = \sqrt{V[\hat{\beta}_0]} = \sqrt{V[\bar{Y}]} = \sqrt{\frac{V[Y]}{n}},$$

which we can estimate via plug-in estimation:

$$\hat{\sigma}[\hat{\beta}_0] = \sqrt{\hat{V}[\hat{\beta}_0]} = \sqrt{\hat{V}[\bar{Y}]} = \sqrt{\frac{\hat{V}[Y]}{n}}.$$

As before, we could use the estimated sampling variance to compute a 95% confidence interval for the population mean, β_0 : $CI_{.95}(\beta_0) = (\hat{\beta}_0 - 1.96\hat{\sigma}[\hat{\beta}_0], \hat{\beta}_0 + 1.96\hat{\sigma}[\hat{\beta}_0])$.

This simple case will provide some intuition going forward. Estimating standard errors becomes more complicated when we have one or more explanatory variables, but the general *principles* for estimating uncertainty, confidence intervals, and *p*-values translate from the sample mean to the regression estimator. For each coefficient β_k , it is possible to derive an estimate of the standard error of $\hat{\beta}_k$:

$$\hat{\sigma}[\hat{\beta}_k] = \sqrt{\hat{V}[\hat{\beta}_k]}.$$

Given the usual plug-in regularity conditions, *robust standard errors* or the bootstrap provide consistent estimates of the asymptotic standard errors under i.i.d. sampling. We consider these in the following two sections.

4.2.2 Inference with Robust Standard Errors and the Bootstrap

In this section, we discuss a consistent estimator for the standard errors of the regression estimators using matrix algebra. Define the vector of errors $\boldsymbol{\varepsilon}$ as the differences between the observed values of Y and the (true) BLP of Y given the observed values of $X_{[1]}, X_{[2]}, \dots, X_{[K]}$:

$$\boldsymbol{\varepsilon} = \mathbf{Y} - \mathbb{X}\boldsymbol{\beta}.$$

We can then decompose $\hat{\boldsymbol{\beta}}$ as

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \mathbf{Y} \\ &= (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T (\mathbb{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \mathbb{X}\boldsymbol{\beta} + (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \boldsymbol{\varepsilon} \\ &= \boldsymbol{\beta} + (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \boldsymbol{\varepsilon}.\end{aligned}$$

This representation allows us to highlight the particular role that the deviations from the BLP play in generating the sampling variability of the regression estimator. Under the usual plug-in regularity conditions, $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ converges in distribution to a multivariate normal random vector⁵ with expectation zero and covariance matrix

$$\left(E[\mathbf{X}^T \mathbf{X}]\right)^{-1} E[\varepsilon^2 \mathbf{X}^T \mathbf{X}] \left(E[\mathbf{X}^T \mathbf{X}]\right)^{-1},$$

⁵ A random vector is said to be *multivariate normal* if every linear combination of its components has a normal distribution.

where $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$. We omit a derivation, but see Powell (2010), Hansen (2013, Ch. 7.2), or the informal treatment in Angrist and Pischke (2009, pp. 40–48), for constructive derivations under mild regularity conditions.

To estimate $V[\hat{\beta}]$, we need only apply the plug-in principle: plug in residuals for errors and sample means for expected values. The resulting plug-in estimator is known as the *robust sampling variance estimator* (or *sandwich estimator*).⁶

Definition 4.2.1. *Robust Sampling Variance Estimator for OLS*

Let $(Y_1, \mathbf{X}_1), (Y_2, \mathbf{X}_2), \dots, (Y_n, \mathbf{X}_n)$ be i.i.d. random vectors, where $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$. Let $\hat{\beta}$ be the OLS coefficient estimator and $e_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_{[1]i} + \hat{\beta}_2 X_{[2]i} + \dots + \hat{\beta}_K X_{[K]i}), \forall i \in \{1, 2, \dots, n\}$. Then the *robust sampling variance estimator* for $\hat{\beta}$ is

$$\begin{aligned}\hat{V}[\hat{\beta}] &= \frac{1}{n} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{X}_i^T \mathbf{X}_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n e_i^2 \mathbf{X}_i^T \mathbf{X}_i \right) \left(\frac{1}{n} \sum_{i=1}^n \mathbf{X}_i^T \mathbf{X}_i \right)^{-1} \\ &= (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \text{diag}(e_1^2, e_2^2, \dots, e_n^2) \mathbb{X} (\mathbb{X}^T \mathbb{X})^{-1},\end{aligned}$$

where $\text{diag}(e_1^2, e_2^2, \dots, e_n^2)$ is the $n \times n$ matrix whose i^{th} diagonal element is e_i^2 and whose off-diagonal elements are all zero.

(Note that $V[\hat{\beta}]$ is a covariance matrix—see Definition 2.3.1.) The estimator is known as “robust” because its validity does not require any assumptions beyond the usual plug-in regularity conditions. (The following section will consider an alternative type of variance estimation that leans on additional assumptions.) Importantly, with the robust sampling variance estimator in hand, we can derive the *robust standard error* for each OLS regression coefficient: the robust standard error estimator for β_k is the square root of the k^{th} diagonal element of $\hat{V}[\hat{\beta}]$.

Recall that the robust standard error is a plug-in estimate of an asymptotic approximation of the standard error, not the true standard error. Without assumptions that would strain credulity (such as normality of errors and fixed explanatory variables), we cannot derive an unbiased estimator of—much less an exact value for—the standard error of our regression estimates given any finite n . However, under the usual plug-in regularity conditions, robust standard errors are consistent.

⁶ It is called the sandwich estimator because the $\text{diag}(e_1^2, e_2^2, \dots, e_n^2)$ term (the “meat”) is sandwiched between the $(\mathbb{X}^T \mathbb{X})^{-1}$ terms (the “bread”).

The bootstrap (Section 3.4.3) is again available as an alternative, and the bootstrap standard errors are also consistent under the usual plug-in regularity conditions. The bootstrap and robust standard errors agree in large samples but, as an exact resampling method, the bootstrap may have better finite sample properties. When in doubt, we recommend using the bootstrap.

Regardless of whether or not we estimate the standard error using robust standard errors or the bootstrap, we use these standard error estimates to compute normal approximation-based confidence intervals and p -values, as before (Sections 3.4.1 and 3.4.2). The following example illustrates the applicability of this approach.

Example 4.2.2. *Normal Approximation-Based Confidence Intervals for Regression*

Suppose we had a random vector $(X_{[1]}, X_{[2]}, Y)$ with the BLP

$$g(X_{[1]}, X_{[2]}) = \beta_0 + \beta_1 X_{[1]} + \beta_2 X_{[2]},$$

and suppose we obtained the following *estimates* after fitting the regression:

$$\hat{\beta}_0 = 0.60, \hat{\sigma}[\hat{\beta}_0] = 0.25,$$

$$\hat{\beta}_1 = 0.75, \hat{\sigma}[\hat{\beta}_1] = 1.25,$$

$$\hat{\beta}_2 = 3.00, \hat{\sigma}[\hat{\beta}_2] = 1.00.$$

We could then construct the following 95% confidence intervals under a normal approximation:

$$CI_{.95}(\beta_0) = (0.11, 1.09),$$

$$CI_{.95}(\beta_1) = (-1.70, 3.20),$$

$$CI_{.95}(\beta_2) = (1.04, 4.96).$$

Then, for instance, we can say approximately with 95% confidence (if n is large and the usual plug-in regularity conditions hold) that the *conditional* slope of the BLP with respect to $X_{[2]}$ lies in the interval $(1.04, 4.96)$.⁷ Likewise, normal approximation-based p -values can be computed in the manner shown in Chapter 3. So, for example, we can state that, under the null hypothesis that β_2 is zero, the probability that we would have observed a $\hat{\beta}_2$ as extreme as we did is less than 0.05. $\triangle$

⁷ In other words, if we repeatedly sampled n observations from $(X_{[1]}, X_{[2]}, Y)$, our interval would cover the true *conditional* slope of the BLP 95% of the time.

4.2.3 Classical Standard Errors

There is an alternative to robust standard errors and the bootstrap, known as *classical standard errors*.

Suppose we *assume* that $E[\varepsilon | X_{[1]}, X_{[2]}, \dots, X_{[K]}] = 0$ and $V[\varepsilon | X_{[1]}, X_{[2]}, \dots, X_{[K]}] = \sigma^2$ for all possible values of $X_{[1]}, X_{[2]}, \dots, X_{[K]}$. The first assumption is equivalent to linearity of $E[Y | X_{[1]}, X_{[2]}, \dots, X_{[K]}]$, and the second assumption is known as (conditional) *homoskedasticity*. Again, let $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$. Then, as before,

$$\begin{aligned} nV[\hat{\boldsymbol{\beta}}] &\approx \left(E[\mathbf{X}^T \mathbf{X}]\right)^{-1} E[\varepsilon^2 \mathbf{X}^T \mathbf{X}] \left(E[\mathbf{X}^T \mathbf{X}]\right)^{-1} \\ &= \left(E[\mathbf{X}^T \mathbf{X}]\right)^{-1} \left(\sigma^2 E[\mathbf{X}^T \mathbf{X}]\right) \left(E[\mathbf{X}^T \mathbf{X}]\right)^{-1} \\ &= \sigma^2 \left(E[\mathbf{X}^T \mathbf{X}]\right)^{-1}, \end{aligned}$$

where the approximation follows from Section 4.2.2, and the first equality follows from the facts that $E[\varepsilon^2 \mathbf{X}^T \mathbf{X}] = E[E[\varepsilon^2 \mathbf{X}^T \mathbf{X} | \mathbf{X}]]$ (by the Law of Iterated Expectations) and $E[\varepsilon^2 | \mathbf{X}] = \sigma^2$ (by the linearity and homoskedasticity assumptions).

We can then derive the plug-in estimator of $V[\hat{\boldsymbol{\beta}}]$, known as the *classical sampling variance estimator*.

Definition 4.2.3. *Classical Sampling Variance Estimator for OLS*

Let $(Y_1, \mathbf{X}_1), (Y_2, \mathbf{X}_2), \dots, (Y_n, \mathbf{X}_n)$ be i.i.d. random vectors, where $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$. Let $\hat{\boldsymbol{\beta}}$ be the OLS coefficient estimator and $e_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_{[1]i} + \hat{\beta}_2 X_{[2]i} + \dots + \hat{\beta}_K X_{[K]i})$, $\forall i \in \{1, 2, \dots, n\}$. Then the *classical sampling variance estimator* for $\hat{\boldsymbol{\beta}}$ is

$$\hat{V}_C[\hat{\boldsymbol{\beta}}] = \frac{1}{n} \hat{\sigma}^2 \left(\frac{1}{n} \sum_{i=1}^n \mathbf{X}_i^T \mathbf{X}_i \right)^{-1} = \hat{\sigma}^2 (\mathbb{X}^T \mathbb{X})^{-1},$$

where $\hat{\sigma}^2 = \hat{V}[Y - \mathbf{X}\hat{\boldsymbol{\beta}}]$.

This is almost the same as the sampling variance estimator for regression given in most textbooks.⁸ The classical standard error estimator for β_k is then just defined as square root of the k^{th} diagonal element of $\hat{V}_C[\hat{\boldsymbol{\beta}}]$.

Remember, however, that classical standard errors rely on a strong assumption: that the magnitude of the deviations from the BLP is *unrelated* to the

⁸ The only difference is that, in most textbooks, $\hat{\sigma}^2$ uses a denominator of $n - (K + 1)$ instead of $n - 1$.

values of the X s. When this assumption and the usual plug-in regularity conditions hold, the classical sampling variance will be consistent. However, this assumption should not be invoked lightly; in real-world problems, where we have no guarantees about the process that gives rise to the data, the errors may be *heteroskedastic*. If homoskedasticity *does* hold, then in large samples, the classical and robust sampling variance estimates will agree. With large n and i.i.d. sampling, there is little reason not to use robust standard errors or the bootstrap to characterize the uncertainty of a regression fit.

4.3 ESTIMATION OF NONLINEAR CONDITIONAL EXPECTATION FUNCTIONS

In this section, we consider estimating nonlinear CEFs by changing the *specification* of the regression. By specification, we mean the manner and functional form that we use to relate explanatory variables to the outcome. We illustrate this idea by returning to an example from Chapter 2, Example 2.2.23 (*Plotting the CEF and the BLP*).

Example 4.3.1. Best Quadratic Predictor

Let X and Y be random variables with $X \sim U(0, 1)$ and $Y = 10X^2 + W$, where $W \sim N(0, 1)$ and $X \perp W$. As shown in Example 2.2.23, the CEF of Y given X is $E[Y|X] = 10X^2$, and the BLP of Y given X is $g(X) = -\frac{5}{3} + 10X$. Figure 4.3.1 plots 1,000 random draws of (X, Y) and then superimposes the graphs of the CEF (in black) and the BLP (in red), as shown in Figure 2.2.1.

Although the CEF is nonlinear, the BLP is a good first approximation of the CEF. Thus, OLS regression, which estimates the BLP, provides a principled first approximation of the CEF. The BLP is a good, but not great, approximation of the CEF. We can, however, do better while still using OLS regression.

Continuing with the above example, suppose we now create a second explanatory variable, X^2 . Then we can use a new regression specification to estimate the best *quadratic* predictor:

$$\beta = \operatorname{argmin}_{(b_0, b_1, b_2) \in \mathbb{R}^3} E[(Y - b_0 - b_1X - b_2X^2)^2].$$

In this example, the best linear predictor of Y given X and X^2 is the CEF:

$$\begin{aligned} g(X, X^2) &= E[Y|X, X^2] \\ &= E[Y|X] \\ &= \beta_0 + \beta_1X + \beta_2X^2 \\ &= 0 + 0X + 10X^2 \\ &= 10X^2. \end{aligned}$$

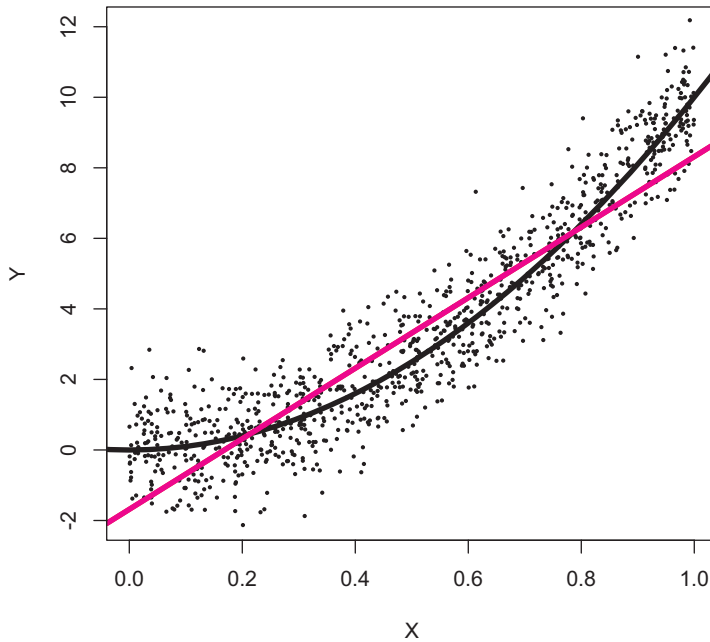


FIGURE 4.3.1. *Plotting the CEF and BLP*

More generally, if the CEF of Y is linear in X and X^2 (in other words, quadratic in X),

$$E[Y|X] = \beta_0 + \beta_1 X + \beta_2 X^2,$$

then the BLP of Y given X and X^2 is the CEF, so regression of Y on X and X^2 consistently estimates the CEF. If the CEF is not linear in X and X^2 , then regression of Y on X and X^2 will approximate the CEF insofar as

$$E[Y|X] \approx \beta_0 + \beta_1 X + \beta_2 X^2.$$

It is important to note that, once we introduce nonlinearity, interpreting the coefficients becomes less straightforward. With bivariate regression, the estimated slope of the CEF of Y with respect to X was $\hat{\beta}_1$. Here, with a nonlinear function, we can take the derivative.

If the CEF is linear in X and X^2 , then the slope is

$$\frac{\partial E[Y|X]}{\partial X} = \frac{\partial (\beta_0 + \beta_1 X + \beta_2 X^2)}{\partial X} = \beta_1 + 2\beta_2 X.$$

The slope of the CEF thus depends on the value of X . So, if we estimated the β s using OLS, our estimate of the slope of the CEF at $X = x$ would be $\hat{\beta}_1 + 2\hat{\beta}_2 x$. Δ

In the following sections, we show that this general idea—changing how we specify the functional form of the predictor—can allow us to make even OLS regression remarkably flexible.

4.3.1 Polynomials

We can provide a more general basis for approximating the CEF using regression. The *Weierstrass Approximation Theorem* states that any continuous function defined on a closed interval can be uniformly approximated to arbitrary precision using a polynomial function.

Theorem 4.3.2. *Weierstrass Approximation Theorem*

Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous. Then, $\forall \epsilon > 0$, there exists a polynomial p such that $\forall x \in [a, b]$, $|f(x) - p(x)| < \epsilon$.

We omit the proof of this theorem. We can use the Weierstrass Approximation Theorem to derive the following result.

Theorem 4.3.3. *Polynomial Approximation of the CEF*

Let X and Y be random variables, and suppose that $E[Y|X = x]$ is continuous and $\text{Supp}[X] = [a, b]$. Then, $\forall \epsilon > 0$, $\exists K \in \mathbb{N}$ such that, $\forall K' \geq K$,

$$E\left[\left(E[Y|X] - g(X, X^2, \dots, X^{K'})\right)^2\right] < \epsilon,$$

where $g(X, X^2, \dots, X^{K'})$ is the BLP of Y given $X, X^2, \dots, X^{K'}$.

Proof: Let $\epsilon > 0$. Then $\sqrt{\epsilon} > 0$, so by the Weierstrass Approximation Theorem, there exists a polynomial p such that $\forall x \in [a, b]$,

$$|E[Y|X = x] - p(x)| < \sqrt{\epsilon}.$$

Thus,

$$E\left[\left(E[Y|X] - p(X)\right)^2\right] < \left(\sqrt{\epsilon}\right)^2 = \epsilon.$$

Let K be the degree of p . The BLP of Y given $X, X^2, \dots, X^K$ is the minimum MSE predictor of $E[Y|X]$ among polynomials of degree less than or equal to K , so by definition

$$E\left[\left(E[Y|X] - g(X, X^2, \dots, X^K)\right)^2\right] \leq E\left[\left(E[Y|X] - p(X)\right)^2\right] < \epsilon,$$

where $g(X, X^2, \dots, X^K)$ is the BLP of Y given $X, X^2, \dots, X^K$. Now, let $K' \geq K$.

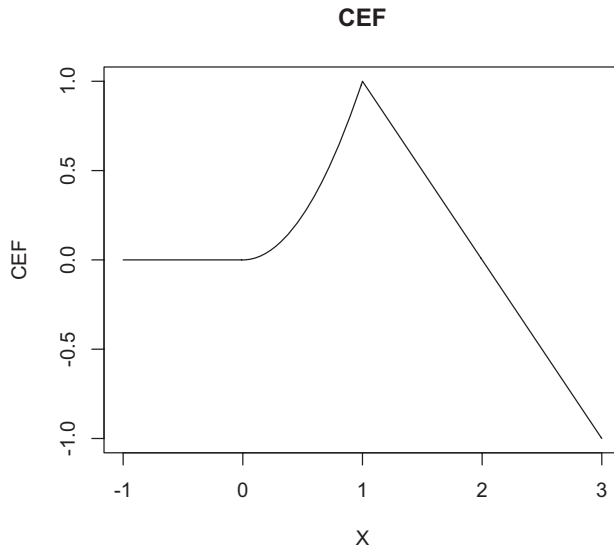


FIGURE 4.3.2. A Nonlinear CEF

The BLP of Y given $X, X^2, \dots, X^{K'}$ is the minimum MSE predictor of $E[Y|X]$ among polynomials of degree less than or equal to K' , so by definition

$$E\left[\left(E[Y|X] - g(X, X^2, \dots, X^{K'})\right)^2\right] \leq E\left[\left(E[Y|X] - g(X, X^2, \dots, X^K)\right)^2\right] < \epsilon. \quad \square$$

Thus, as long as the CEF is continuous and the support of X is a closed interval, we can get MSE as small as we want with a polynomial BLP of sufficiently high degree.

Let us visualize this. Consider the following CEF, plotted in Figure 4.3.2:

$$E[Y|X=x] = \begin{cases} 0 & : x \in [-1, 0] \\ x^2 & : x \in (0, 1) \\ 2-x & : x \in [1, 3]. \end{cases}$$

We can approximate this CEF with polynomials of X . Figure 4.3.3 shows the BLPs (assuming that $X \sim U(-1, 3)$) using polynomials of degrees 1, 2, 3, 4, 5, and 10.

This is *not* to suggest that it is good practice to include a tenth-order polynomial in our regression (see Section 4.3.2). But the point stands: we can approximate the CEF to arbitrary precision using a BLP, as long as the CEF is continuous. Additionally, since regression is consistent for the BLP, this means that with large enough n , we can estimate the CEF to arbitrary precision. (More on this in Section 4.3.5, *Sieve Estimation*.)

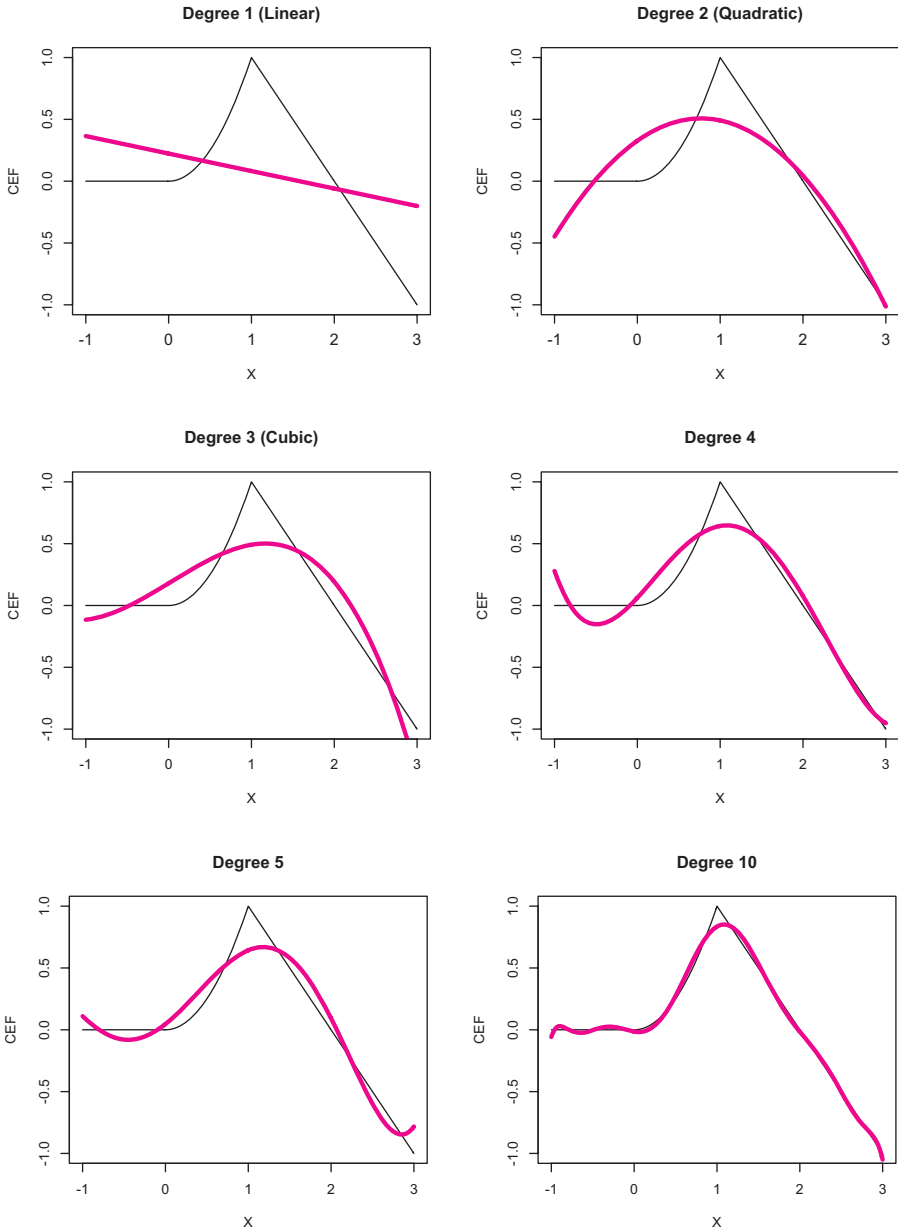


FIGURE 4.3.3. Polynomial Approximations of a Nonlinear CEF

The linearity assumption of regression can be relaxed in many ways; polynomials are just one way. (High-order polynomials are not a particularly good way, for reasons beyond the scope of this book—see Runge 1901; Hansen 2013, Ch. 12.6.) The remainder of this chapter describes some other methods, but before we proceed, we first discuss the key challenge encountered in doing so.

4.3.2 Overfitting

Although including many higher-order terms in a regression relaxes the assumption of linearity, it may introduce problems. In particular, using a complex specification can lead to the problem of *overfitting*. Put simply, the data suffer from micronumerosity—we do not have enough observations to fit a complex approximation well. When n is small, estimating a large number of regression coefficients may lead us to be highly sensitive to the “noise” in the data.⁹ As we demonstrate in the following example, the problem of overfitting is clearest in the case where the CEF is, in fact, linear.

Example 4.3.4. *Overfitting with a Linear CEF*

Suppose we had a random vector (X, Y) such that

$$Y = X + 5 + \varepsilon,$$

where $X \sim U(-10, 10)$,¹⁰ $\varepsilon \sim N(0, 25)$, and $X \perp \varepsilon$. The CEF of Y given X is thus $E[Y|X = x] = x + 5$.

We can compare what might happen if we tried to estimate the CEF using a linear regression versus regression with a fifth-order polynomial. First, suppose we had just ten i.i.d. observations of (X, Y) . Panel (a) of Figure 4.3.4 plots ten random draws of (X, Y) . In black is the true CEF (which, since it is linear, is also the true BLP). In red is the linear regression estimate given this data. Panel (b) displays the true CEF in black again, but the fifth-order polynomial regression estimate is in red.

With $n = 10$, the fifth-order polynomial does not appear to perform well as an approximation of the true CEF, since it effectively “over interprets” the random noise in the data as indicative of the underlying relationship between X and Y . This is why we call it overfitting: with small n , the more flexible

⁹ Overfitting can also occur when researchers fit multiple specifications to the data and report the best fitting specification; this can be shown to be equivalent to estimating additional parameters in terms of introducing overfitting. In our discussion, we assume that the inferential procedure is fixed ex ante. In practice, this assumption can be approximated through “sample splitting,” which allows researchers to control overfitting by using different subsamples of the data for estimation as for inference. The logic of this procedure can be used to control overfitting more generally, as we will see when we briefly discuss a related procedure, cross-validation, in Section 4.3.6 (*Penalized Regression*).

¹⁰ In other words, X follows a *uniform distribution* with constant density on the interval $[-10, 10]$ (and zero elsewhere).

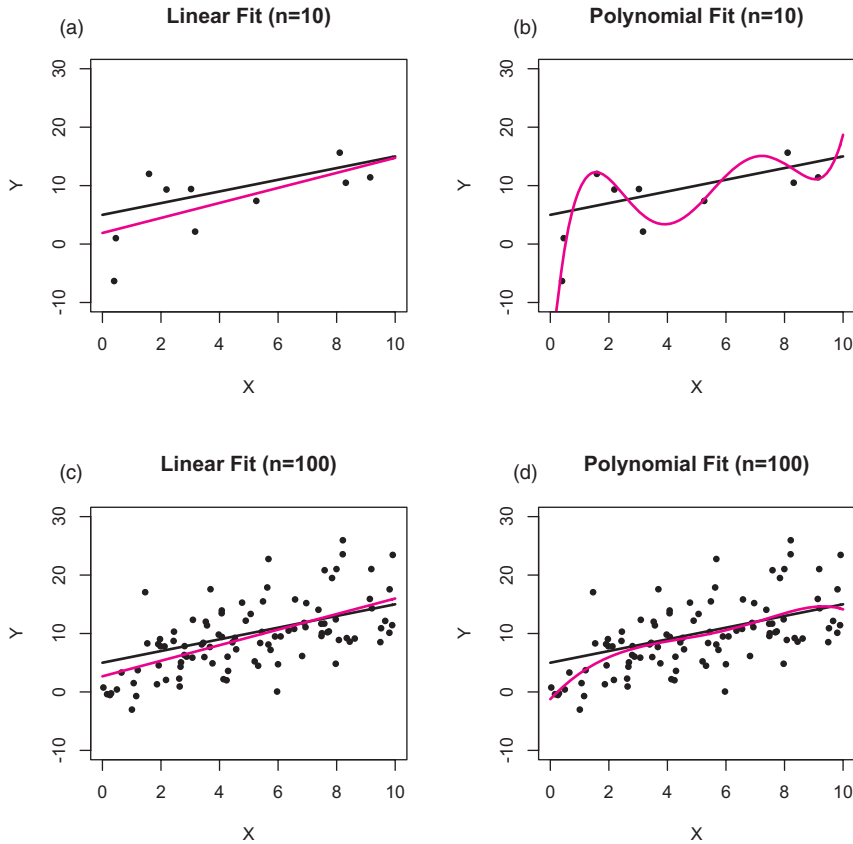
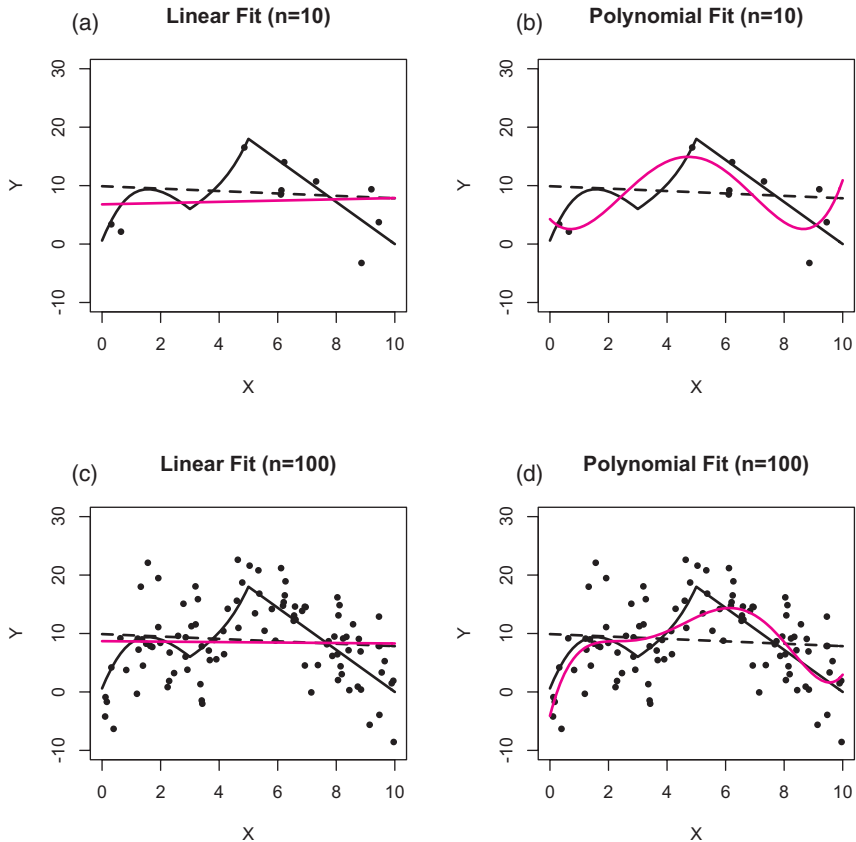


FIGURE 4.3.4. *Overfitting with a Linear CEF*

regression specification fits the estimate to the observed data *too well*. By contrast, the linear regression estimate approximates the true CEF fairly well considering the small sample size.

As overfitting is simply a case of micronumerosity (Section 3.4.4) in estimating our complicated approximation, it goes away when we have a large enough sample size. With only ten observations, trying to estimate a fifth-order polynomial (six coefficients) means we have less than two observations per coefficient to be estimated. If we went even further and estimated a ninth-order polynomial (ten coefficients), the resulting curve would pass exactly through every single observation. We would essentially be treating each observation as an $n = 1$ estimate of the CEF at that value of X . Panels (c) and (d) of Figure 4.3.4 plot 100 random draws of (X, Y) and again superimposes the true CEF (in black), as well as the linear regression fit and the fifth-order

FIGURE 4.3.5. *Overfitting with a Nonlinear CEF*

polynomial regression fits (in red). With $n = 100$, both regression estimates approximate the true CEF. Δ

But what if the CEF actually *is* highly nonlinear? Even then, a more flexible specification is not necessarily better. The problem of overfitting remains.

Example 4.3.5. *Overfitting with a Nonlinear CEF*

Suppose we had a random vector (X, Y) such that

$$Y = g(X) + \varepsilon,$$

for some (nonlinear) function g , where $X \sim U(-10, 10)$, $\varepsilon \sim N(0, 25)$, and $X \perp \varepsilon$. The CEF of Y given X is thus $E[Y|X = x] = g(X)$. Let g be the function shown in black in Figure 4.3.4.

As in Figure 4.3.4, panel (a) Figure 4.3.5 plots ten random draws of (X, Y) and superimposes the true CEF (in black) and the linear

regression fit (in red). Panel (b) presents the same, except with the fifth-order polynomial regression fit (in red). In both, the true BLP is illustrated with a dotted black line. As before, with $n = 10$, the fifth-order polynomial regression overfits badly. The linear regression estimate, on the other hand, is close to the BLP, and thus, by definition, nearly coincides with the best possible linear approximation of the CEF (in terms of MSE).

Once again, with more data, we can afford to be “greedier” in trying to approximate the shape of the CEF in greater detail. Panels (c) and (d) of Figure 4.3.5 plot 100 random draws of (X, Y) . With $n = 100$, the fifth-order polynomial regression approximates the true CEF quite well. The linear regression, though, gives an even more precise estimate of the BLP. Even with large n and a highly nonlinear true CEF, linear regression still has the virtue of being a safe, conservative estimator that yields a good approximation of a meaningful summary feature of the joint distribution. $\triangle$

4.3.3 Interactions

We have shown that, with polynomials, we can approximate any continuous CEF in one variable to arbitrary precision. This was, in a sense, a proof of concept. The same principle generalizes to the multivariate case. To see this, consider the case where we are trying to predict Y using $X_{[1]}$ and $X_{[2]}$, and the slope of the CEF with respect to $X_{[1]}$ depends on the value of $X_{[2]}$. The BLP is still the BLP, but it might fail to capture an interesting *interaction* between the variables $X_{[1]}$ and $X_{[2]}$. To formalize a bit, suppose the CEF is otherwise linear, but includes an *interaction term* $X_{[1]}X_{[2]}$:

$$E[Y | X_{[1]}, X_{[2]}] = \beta_0 + \beta_1 X_{[1]} + \beta_2 X_{[2]} + \beta_3 X_{[1]}X_{[2]}.$$

Much as before, we can derive the slope of the CEF with respect to $X_{[1]}$ and $X_{[2]}$ by taking partial derivatives:

$$\begin{aligned} \frac{\partial E[Y | X_{[1]}, X_{[2]}]}{\partial X_{[1]}} &= \frac{\partial (\beta_0 + \beta_1 X_{[1]} + \beta_2 X_{[2]} + \beta_3 X_{[1]}X_{[2]})}{\partial X_{[1]}} = \beta_1 + \beta_3 X_{[2]}, \\ \frac{\partial E[Y | X_{[1]}, X_{[2]}]}{\partial X_{[2]}} &= \frac{\partial (\beta_0 + \beta_1 X_{[1]} + \beta_2 X_{[2]} + \beta_3 X_{[1]}X_{[2]})}{\partial X_{[2]}} = \beta_2 + \beta_3 X_{[1]}. \end{aligned}$$

Thus, the *conditional* slope of the CEF with respect to $X_{[1]}$ depends on the value of $X_{[2]}$, and vice versa. In this setting (unlike in Section 4.3.1, *Polynomials*), each coefficient has a clear interpretation. If the CEF is linear in $X_{[1]}$, $X_{[2]}$, and $X_{[1]}X_{[2]}$, then

- $\beta_0 = E[Y | X_{[1]} = 0, X_{[2]} = 0]$.
- β_1 is the slope of the CEF with respect to $X_{[1]}$ when $X_{[2]} = 0$.
- β_2 is the slope of the CEF with respect to $X_{[2]}$ when $X_{[1]} = 0$.

- β_3 is how each of the slopes changes when the value of the other variable is increased by one. (For example, if $X_{[1]}$ moves from 1 to 2, then the slope of the CEF with respect to $X_{[2]}$ increases by β_3 ; if $X_{[2]}$ moves from 7 to 4, then the slope of the CEF with respect to $X_{[1]}$ decreases by $3\beta_3$.)

Of course, if $\beta_3 = 0$, then $E[Y|X_{[1]}, X_{[2]}] = \beta_0 + \beta_1 X_{[1]} + \beta_2 X_{[2]}$ and there is no interaction. All of this can be generalized for any number of X variables.

For even greater flexibility, interactions can be combined with higher-order polynomial terms.¹¹ For example,

$$E[Y|X_{[1]}, X_{[2]}] = \beta_0 + \beta_1 X_{[1]} + \beta_2 X_{[2]} + \beta_3 X_{[1]}X_{[2]} + \beta_4 X_{[1]}^2 + \beta_5 X_{[2]}^2 \\ + \beta_6 X_{[1]}^2 X_{[1]} + \beta_7 X_{[1]} X_{[2]}^2 + \beta_8 X_{[1]}^2 X_{[2]}^2.$$

Once higher-order terms are included, the coefficients no longer have straightforward interpretations. Conditional slopes with respect to each X variable can still be computed by taking partial derivatives. Beyond that, it is usually easiest to compute the implied value of $E[Y|X_{[1]}, X_{[2]}]$ for different values of $X_{[1]}$ and $X_{[2]}$ to see how the conditional expectation of Y changes as $X_{[1]}$ and $X_{[2]}$ change.

Again, OLS regression yields a consistent estimate of the β s for any of the above specifications. When the CEF is not linear in the variables specified, OLS consistently estimates the BLP of Y given these variables, that is, the best approximation of the CEF among all functions of the form specified by the researcher.

4.3.4 Summarizing Partial Derivatives of the CEF

The previous sections on polynomials and interactions described how we can characterize partial derivatives of the CEF, even when we are considering a nonlinear CEF. These partial derivatives of the CEF are referred to as *marginal effects* in the econometrics literature, but we refrain from using the word “effect” outside the domain of a causal model (as in Chapter 7).¹² In this section, we consider how to summarize these partial derivatives, instead of just considering them at one particular value.

¹¹ Theorem 4.3.3 (*Polynomial Approximation of the CEF*) can be generalized to multiple explanatory variables: if we include enough polynomial and interaction terms, we can approximate the CEF to arbitrary precision, provided the CEF is continuous and the support of $\mathbf{X}$ is a compact set.

¹² If our readers are so inclined, they may mentally replace every future instance of the term “partial derivative” in this section with “marginal effect,” but they should bear in mind that these do not necessarily represent causal effects without further assumptions.

Let us assume that the CEF is differentiable with respect to a given variable $X_{[k]}$.¹³ Then, the partial derivative of the CEF with respect to $X_{[k]}$ is

$$\left. \frac{\partial E[Y|X]}{\partial X_{[k]}} \right|_{X=x}.$$

Again, a plug-in estimator using the estimated CEF,

$$\left. \frac{\partial \hat{E}[Y|X]}{\partial X_{[k]}} \right|_{X=x},$$

will be consistent under proper specification and the usual plug-in regularity conditions.

Thus, it is straightforward to characterize and estimate the partial derivative of CEF of Y given X . We now consider how to summarize the relationship between Y and a given variable $X_{[k]}$. The two most common quantities are the *average partial derivative* and the *partial derivative at the average*. We consider these in turn.

The average partial derivative represents how much a change in a particular variable $X_{[k]}$ moves the CEF, taking the expectation over the entire distribution of (Y, X) . In this sense, it is the average slope of the CEF with respect to $X_{[k]}$.

Definition 4.3.6. Average Partial Derivative

For a random variable Y and random vector X , the *average partial derivative* of the CEF (of Y given X) with respect to $X_{[k]}$ is

$$APD_{X_{[k]}} = E \left[\frac{\partial E[Y|X]}{\partial X_{[k]}} \right].$$

A natural plug-in estimator for $APD_{X_{[k]}}$ is obtained by replacing the expected value with a sample mean and the CEF with the estimated CEF:

$$\widehat{APD}_{X_{[k]}} = \frac{1}{n} \sum_{i=1}^n \left(\left. \frac{\partial \hat{E}[Y|X]}{\partial X_{[k]}} \right|_{X=X_i} \right).$$

Put simply, $\widehat{APD}_{X_{[k]}}$ entails estimating the partial derivative for every observation, and taking an average across all observations. If the regression specification coincides with the CEF, then $\widehat{APD}_{X_{[k]}}$ is a consistent estimator of $APD_{X_{[k]}}$.

¹³ Beyond requiring that the CEF be differentiable at every value of X with positive support, this characterization only makes sense when X is continuous. When X is discrete, we would recommend computing variants of discrete analogs to partial derivatives, such as partial slopes.

The partial derivative at the average represents a conceptually different target: how much a change in a particular variable $X_{[k]}$ moves the CEF when all variables in $\mathbf{X}$ are equal to their expected value. It is the slope of the CEF with respect to $X_{[k]}$ at a single point.

Definition 4.3.7. Partial Derivative at the Average

For a random variable Y and random vector $\mathbf{X}$, the *partial derivative at the average* of the CEF (of Y given $\mathbf{X}$) with respect to $X_{[k]}$ is

$$PDA_{X_{[k]}} = \left. \frac{\partial E[Y|\mathbf{X}]}{\partial X_{[k]}} \right|_{\mathbf{X}=E[\mathbf{X}]}.$$

Again, a natural plug-in estimator is available for $PDA_{X_{[k]}}$, which replaces the expected values with sample means, and the CEF with the estimated CEF. The plug-in estimate of the partial derivative at the average with respect to $X_{[k]}$ is

$$\widehat{PDA}_{X_{[k]}} = \left. \frac{\partial \hat{E}[Y|\mathbf{X}]}{\partial X_{[k]}} \right|_{\mathbf{X}=\bar{\mathbf{X}}}.$$

Note that $\widehat{PDA}_{X_{[k]}}$ is easier to compute than $\widehat{APD}_{X_{[k]}}$. $\widehat{PDA}_{X_{[k]}}$ only requires computing the partial derivative at a single point, whereas $\widehat{APD}_{X_{[k]}}$ requires computing the partial derivative at all n points in the data. As the following example shows, $\widehat{APD}_{X_{[k]}}$ is not generally equal to $\widehat{PDA}_{X_{[k]}}$.

Example 4.3.8. Differences between the Average Partial Derivative and the Partial Derivative at the Average

Suppose $X_{[1]} \perp X_{[2]}$ with $X_{[1]} \sim U(0, 1)$ and $X_{[2]} \sim U(0, 1)$. Further suppose that the CEF is

$$E[Y|X_{[1]}, X_{[2]}] = \begin{cases} -X_{[1]} & : X_{[2]} \leq .8 \\ 9X_{[1]} & : X_{[2]} > .8. \end{cases}$$

Then, by the Law of Iterated Expectations,

$$\begin{aligned} \widehat{APD}_{X_{[1]}} &= \frac{\partial(-X_{[1]})}{\partial X_{[1]}} \Pr[X_{[2]} \leq .8] + \frac{\partial(9X_{[1]})}{\partial X_{[1]}} \Pr[X_{[2]} > .8] \\ &= -1 \times .8 + 9 \times .2 = 1. \end{aligned}$$

But the partial derivative at the average gives a very different result:

$$PDA_{X_{[1]}} = \frac{\partial(-X_{[1]})}{\partial X_{[1]}} = -1.$$

While the partial derivative with respect to $X_{[1]}$ is positive on average, when evaluated at the average ($X_{[1]} = X_{[2]} = 0.5$), the partial derivative is negative. Δ

The average partial derivative is an intuitive way to summarize the role of a particular variable in predicting Y —it represents how much, on average, a variable influences the CEF. While the partial derivative at the average also has a well-defined meaning, it is limited to just one particular point on the probability distribution of the explanatory variables. Thus, for a researcher interested in characterizing the overall relationship between Y and $X_{[k]}$, the average partial derivative is typically preferred to the partial derivative at the average.

4.3.5 Sieve Estimation

Recalling that overfitting is a consequence of micronumerosity, we would be more willing to accept a complicated regression specification if we had considerably more data. In other words, we might be willing to accept restrictive specifications with small n , keeping in mind that if we had larger n , we would increase the complexity. This idea is formalized in the notion of *sieve estimation*. The term *sieve estimator* is something of a misnomer; a sieve estimator is not a different type of estimator, but rather an algorithm for determining what estimator we would use for any given n . We can then characterize the asymptotic behavior of the sequence of estimators defined by the algorithm.

Intrinsically, estimating something like a CEF when there is at least one continuous explanatory variable is an infinite-dimensional problem. We might need an infinite number of parameters to fully characterize the CEF—it is not as though we can just assume that it follows a given functional form. Essentially, every dataset is micronumerous if we are trying to estimate an infinitely complex function. All estimators are limited to having just so many “moving parts” given any particular n . But we want to know that if we had very large n , we would obtain an approximation with arbitrary precision.

With sieve estimation, the asymptotics of the estimator are defined to allow for increasing flexibility, such that, as n grows, the estimator becomes more and more flexible. As long as flexibility grows “slowly” enough relative to n , we can show that the estimator will be consistent. We consider here one illustrative case. Suppose we wanted to estimate the CEF of Y given X using OLS with polynomials, as in Section 4.3.1 (*Polynomials*). We can set up an estimator whose complexity depends on n .

Definition 4.3.9. *Polynomial Least Squares Sieve Estimator*

For i.i.d. random vectors $(Y_1, X_1), (Y_2, X_2), \dots, (Y_n, X_n)$, the *polynomial least squares sieve estimator* of the conditional expectation function is

$$\hat{E}_n[Y|X=x] = \sum_{k=0}^{J_n} \hat{\beta}_k x^k,$$

where

$$\hat{\beta} = \arg \min_{\mathbf{b} \in \mathbb{R}^{J_n+1}} \sum_{i=1}^n \left(Y_i - \sum_{k=0}^{J_n} b_k X_i^k \right)^2,$$

where $J_n \rightarrow \infty$ and $\frac{1}{n}J_n \rightarrow 0$.

In other words, we are regressing Y on a polynomial expansion of X , but the degree of that polynomial depends on n . As $n \rightarrow \infty$, the degree of that polynomial $J_n \rightarrow \infty$, but at a slower rate. If n is very small, we estimate the best linear predictor. If n is somewhat larger, we estimate the best quadratic predictor. And so on. As long as the CEF is smooth with appropriate support across X , then $\hat{E}[Y|X=x]$ will be consistent across regions of the data with nonzero probability mass (by the logic of Theorem 4.3.3, *Polynomial Approximation of the CEF*). The logic of sieve estimation provides us some reassurance that, with enough data, we can obtain good approximations of even a complex CEF.

4.3.6 Penalized Regression

In the previous sections, we discussed some natural ways to increase the complexity of the approximation that we are using: interactions and polynomials. In this section, we consider one way to effectively *reduce* the complexity of approximation: *penalization*. In the case of OLS regression, we penalize by introducing a “penalty term” that is proportional to the complexity of the model (by some definition), and minimizing the sum of the penalty term and mean squared residuals. This helps to control overfitting (see Section 4.3.2, *Overfitting*), without forcing the researcher to arbitrarily choose a simpler specification.

Perhaps the best known approach here is the *lasso* (least absolute shrinkage and selection operator, Tibshirani 1996), a form of penalized regression, which provides a way of flexibly estimating the CEF when we have either (1) a large number of variables or (2) a large number of interactions or polynomials (the two scenarios are functionally equivalent). The key for the lasso is that the penalty term is the sum of the absolute value of estimated regression coefficients (typically excluding the intercept).

Definition 4.3.10. *Lasso Estimator*

For i.i.d. random vectors $(Y_1, X_1), (Y_2, X_2), \dots, (Y_n, X_n)$ where $X = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$, the *lasso estimator* with *penalty parameter* λ is

$$\hat{\beta}_\lambda = \arg \min_{\mathbf{b} \in \mathbb{R}^K} \left(\frac{1}{n} \sum_{i=1}^n (Y_i - \mathbf{X}_i \mathbf{b})^2 + \lambda \|\mathbf{b}\|_1 \right),$$

where $\|\cdot\|_1$ is the ℓ^1 norm, that is, $\|\mathbf{b}\|_1 = \sum_{k=1}^K |b_k|$ (the sum of the absolute value of the coefficients).

The interpretation of lasso estimates is as we would expect: $\hat{\beta}$ characterizes an estimate of the CEF (given the functional form of the specification being used). But note that the lasso estimator depends on what value we set λ to. If we let $\lambda = 0$, then the lasso is equivalent to regular OLS regression. If λ is large, the lasso shrinks each element of $\hat{\beta}$ to (or toward) zero unless it substantially helps to reduce the mean of squared residuals. The lasso thus presents a way to use many explanatory variables to “automatically” obtain good predictions—including when we have more variables than observations. If we assume that, as $n \rightarrow \infty$, $\lambda \rightarrow 0$, K remains fixed, and the CEF is linear in $\mathbf{X}$ (as parameterized), then the lasso will be consistent for the CEF under the assumptions otherwise used in this chapter.

In practice, choosing λ is not easy—a great deal of machine learning literature is devoted to figuring out how to choose λ . Similar problems emerge in general for overfitting problems, much as we might struggle to decide whether or not to fit the data with or without a polynomial, or with or without an interaction. One way to address this problem is by using *leave-one-out cross-validation*. In this case, we let the leave-one-out λ , $\lambda_{loo} = \arg \min_{l \in \mathbb{R}^+} [\frac{1}{n} \sum_{i=1}^n (Y_i - \mathbf{X}_i \hat{\beta}_{-i,l})^2]$, where $\hat{\beta}_{-i,l}$ is computed from the lasso procedure with unit i excluded and $\lambda = l$. This often has good operating characteristics, though proving its asymptotic properties is outside the scope of this book. Note that there are many other penalized regression techniques, including, most notably, ridge (ℓ^2 -penalized) regression or penalized regression trees. These methods operate under the same basic principles, sometimes allowing for nonlinearity in different ways. We will return to the notion of penalization in the following chapter, where it serves a similar role in reducing the complexity of our approximation.

4.4 APPLICATION: ACCESS TO CLEAN WATER AND INFANT MORTALITY

We now consider an application of regression using the *Quality of Governance* data (Teorell et al. 2016), a dataset that compiles country-level information from a variety of sources. Our goal is to assess how access to clean water predicts rates of infant mortality at the country level. Infant mortality rate is measured as the number of infants who died before reaching one year of age per 1,000 live births in a given year, and access to clean water is measured as the percentage of the population with access to a source of clean drinking

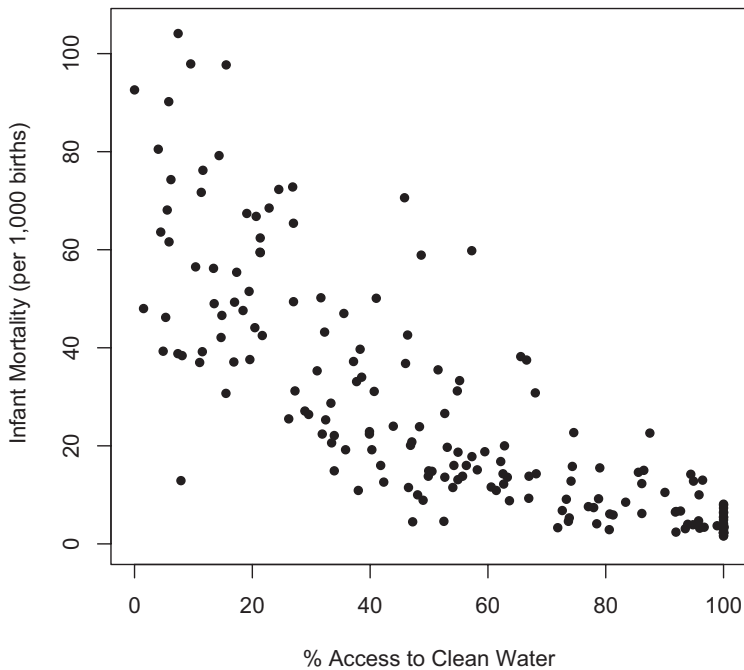
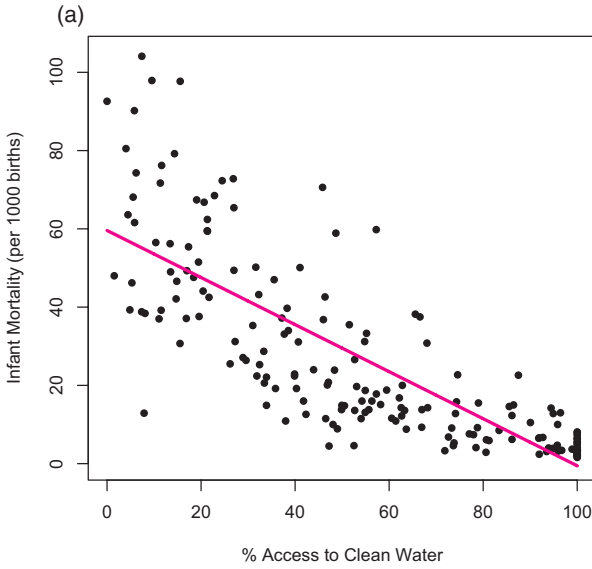


FIGURE 4.4.1. *Plotting Cross-National Infant Mortality Against Access to Clean Water*

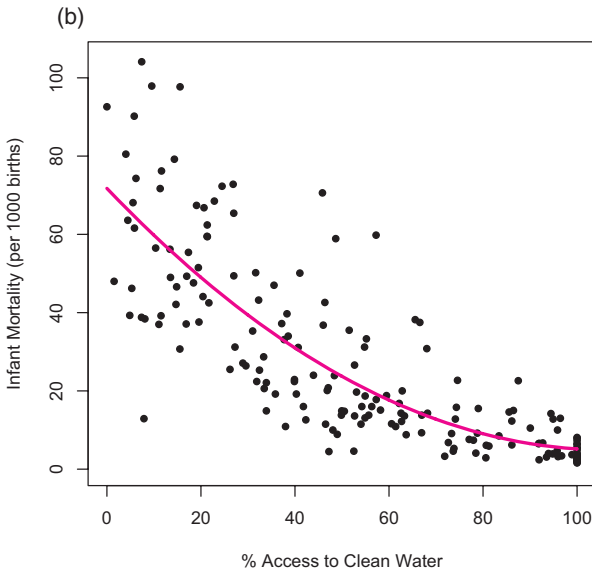
water. We set aside the issue of measurement error here, so we should read all the results in this section as describing the relationships between the *measured* values of country-level variables. It is solely for notational convenience that we omit the qualifier “measured.” The unit of observation is the country, and we have this data for 184 countries, so $n = 184$.

Figure 4.4.1 plots infant mortality against access to clean water. We assume that the data are i.i.d. draws of (Y, X) , where Y denotes infant mortality and X denotes access to clean water. We can use regression to summarize the relationship between these variables. Figure 4.4.2(a) illustrates regression (1), which estimates the best predictor of the CEF of the form $\alpha + \beta X$ (in other words, the BLP). Figure 4.4.2(b) shows a second-order polynomial regression, regression (2), which estimates the best predictor of the CEF of the form $\alpha + \sum_{k=1}^2 \beta_k X^k$. The associated estimates of the coefficients are reported in columns (1) and (2) of Table 4.4.5, with bootstrap standard error estimates in parentheses below each coefficient. The final two rows report estimates of both the average partial derivative (APD) and partial derivative at the average (PDA) with respect to water access.

As one might expect, as access to clean water increases, we see an associated decline in infant mortality. Under specification (1), we estimate that the slope



(1) OLS: $\alpha + \beta X$



(2) OLS: $\alpha + \sum_{k=1}^2 \beta_k X^k$

FIGURE 4.4.2. *Estimates of the CEF of Infant Mortality with Respect to Access to Clean Water*

TABLE 4.4.5. Regression Table for Cross-National Infant Mortality

	Infant Mortality							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Intercept	59.583 (2.803)	71.751 (4.714)	74.186 (3.691)	68.113 (7.559)	76.686 (6.313)	67.385 (18.739)	50.900 (27.180)	68.631 –
Water	–0.601 (0.035)	–1.256 (0.174)	–0.350 (0.038)	–0.706 (0.198)	–0.467 (0.192)	–0.432 (1.190)	1.808 (3.081)	–0.529 –
Water ²	–	0.006 (0.001)	–	0.003 (0.001)	–	–0.006 (0.021)	–0.076 (0.119)	0.000 –
Water ³	–	–	–	–	–	–	0.001 (0.001)	0.000 –
Elec	–	–	–0.367 (0.052)	0.157 (0.274)	–0.398 (0.076)	0.137 (0.855)	0.990 (1.427)	0.062 –
Elec ²	–	–	–	–0.004 (0.002)	–	–0.004 (0.007)	–0.010 (0.013)	–0.003 –
Elec × Water	–	–	–	–	0.001 (0.002)	–0.003 (0.043)	–0.110 (0.113)	0.000 –
Elec × Water ²	–	–	–	–	–	1.80e-4 (5.82e-4)	0.003 (0.003)	8.95e-6 –
Elec × Water ³	–	–	–	–	–	–	–2.22e-5 (2.86e-5)	8.15e-8 –
Elec ² × Water	–	–	–	–	–	1.19e-5 (3.42e-4)	8.20e-4 (9.46e-4)	0.000 –
Elec ² × Water ²	–	–	–	–	–	–9.53e-7 (4.10e-6)	–2.39e-5 (2.26e-5)	0.000 –
Elec ² × Water ³	–	–	–	–	–	–	1.66e-7 (1.78e-7)	0.000 –
$\widehat{APD}_{\text{Water}}$	–0.601 (0.036)	–0.603 (0.043)	–0.350 (0.037)	–0.363 (0.050)	–0.367 (0.052)	–0.350 (0.076)	–0.362 (0.124)	–0.345 (0.075)
$\widehat{PDA}_{\text{Water}}$	–0.601 (0.034)	–0.603 (0.045)	–0.350 (0.038)	–0.363 (0.049)	–0.367 (0.053)	–0.344 (0.140)	–0.092 (0.212)	–0.394 (0.087)

Regressions (1)–(7) report OLS coefficients and regression (8) reports lasso coefficients. $n = 184$ for all regressions. The final two rows report the estimated average partial derivatives and partial derivatives at the average with respect to access to clean water. Bootstrap standard error estimates are reported in parentheses under each estimate.

of the BLP is approximately -0.601 , so for every additional percentage point increase in population with access to clean water, we would predict approximately six fewer deaths before one year of age per 10,000 live births. (Note, however, that there is nothing necessarily causal about this relationship.) The relationship is also nonlinear: the coefficient on the quadratic term in regression (2) is highly statistically significant ($p < 0.001$). But while the CEF is better approximated by the polynomial specification, the key insight is conveyed by

the BLP. To see this, we can note that the APD and PDA estimates are nearly identical across both specifications.

What if we want to consider an additional explanatory variable? We might seek to characterize the relationship between infant mortality and access to clean water, *conditional* on another explanatory variable. We now assume that the data are i.i.d. draws of (Y, X, Z) , where Z is the percentage of the population with access to electricity. Figure 4.4.3 plots infant mortality against access to clean water and electricity.

With two explanatory variables, the estimated approximation of the CEF is no longer a line or curve, but rather a surface. Figure 4.4.4 plots the estimates for six regression specifications. The estimate of the CEF in regression (3) predicts the CEF as a plane. With no higher-order terms or interactions, the coefficients can easily be interpreted as conditional slopes. For example, holding access to electricity constant, for every additional percentage point increase in population with access to clean water, we would predict approximately 3.5 fewer deaths before one year of age per 10,000 live births. Notice that this conditional relationship is smaller than in regression (1). Since we have no interactions, our estimate of the slope of the CEF with respect to X does not depend on the value of Z .

The estimate of the CEF in regression (4) includes quadratic terms for both X and Z , so its graph is a curved surface. Regression (5) includes an interaction term, allowing the conditional slope of the estimated CEF with respect to X to vary depending on the value of Z , and vice versa. However, the coefficient on the interaction turns out to be quite small (and not statistically significant), so the regression (5) estimate is very similar to the regression (3) estimate: the conditional slope with respect to clean water access only slightly increases (that is, becomes less steep) as access to electricity increases.

Regression (6) includes higher-order interaction terms: in addition to an interaction term for X and Z , the specification accounts for the interaction of X and Z^2 , Z and X^2 , and X^2 and Z^2 . Regression (7) includes even more higher-order interaction terms. Finally, regression (8) estimates the CEF using the lasso with λ chosen via leave-one-out cross-validation, with interactions on electricity up to the second order and water up to the third order. When we include higher-order terms—as in regression (4) as well as (6), (7), and (8)—the conditional relationship between clean water access and infant mortality becomes more difficult to characterize from the coefficients alone.

The coefficient estimates for each of these regressions are reported in columns (3) through (8) of Table 4.4.5, with bootstrap standard error estimates in parentheses below each coefficient.¹⁴

¹⁴ Note that the lasso has a “sparsity” property that is manifest here—most coefficients are set to exactly zero. This sparsity property implies that standard errors cannot generally be computed for the lasso coefficients without very strong assumptions, though this is a topic of ongoing inquiry. See Leeb and Pötscher (2006).

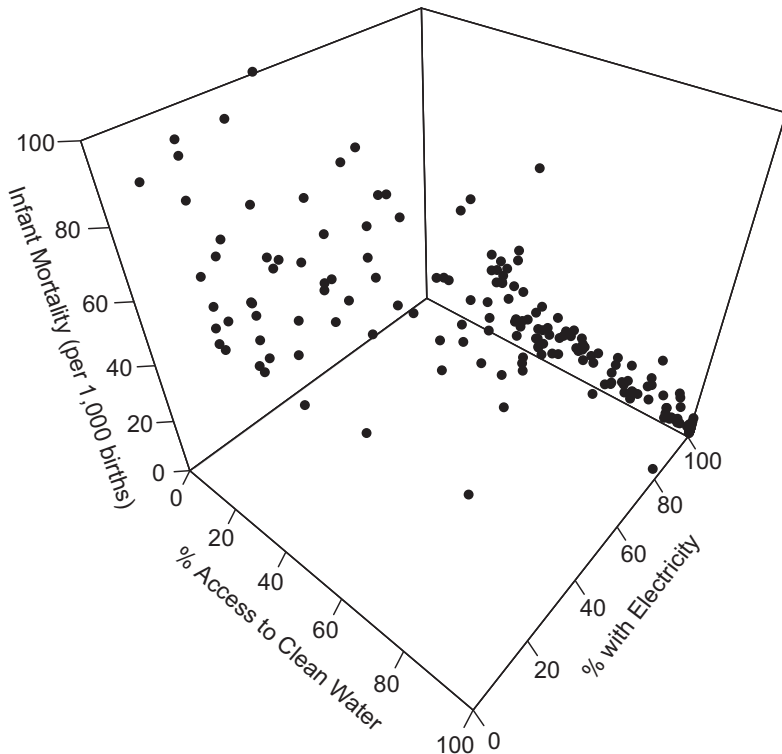


FIGURE 4.4.3. *Plotting Cross-National Infant Mortality Against Access to Clean Water and Electricity*

More generally, we can see how the APD and PDA allow us to draw comparisons across specifications. Once we start conditioning on access to electricity in column (3), we see that our estimates of the APD are nearly halved. Importantly, we see that in columns (3) through (8), our estimate of the APD is largely unchanged by introducing nonlinear specifications, suggesting that the linear approximation may be sufficient for estimating the APD. However, as we increase the complexity of the specification, the standard errors grow quickly for both the estimated APD and PDA, suggesting overfitting. We see that the PDA can be more sensitive to model overfitting—the standard errors are typically larger, and the behavior of the estimator can be sensitive to specification. This is because the PDA is evaluated at a single point, and thus can be very sensitive to nonlinearity, whereas the APD averages across all values. The APD is typically a more sensible way to summarize the conditional relationship between access to water and infant mortality.

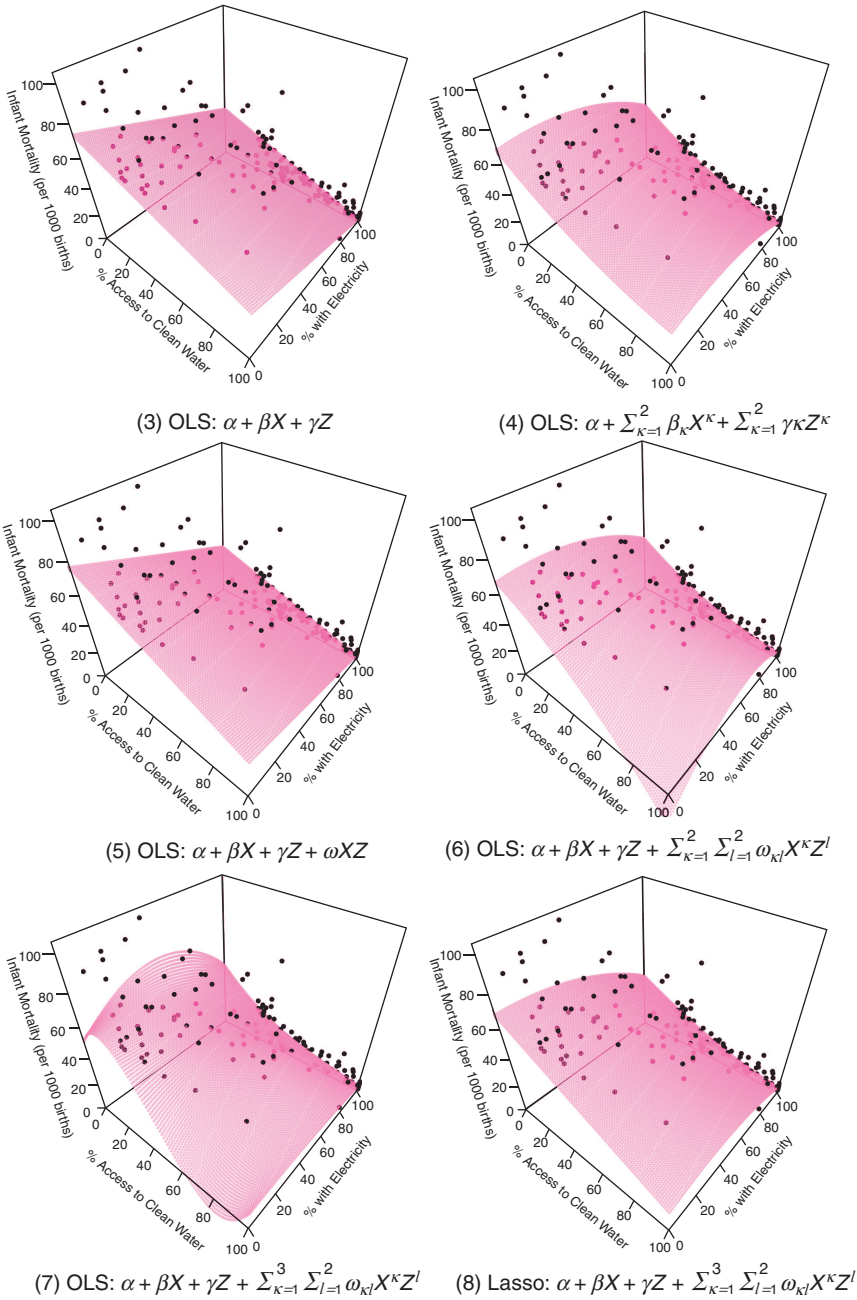


FIGURE 4.4.4. Estimates of the CEF of Infant Mortality with Respect to Access to Clean Water and Electricity

4.5 FURTHER READINGS

Textbook discussions of regression are extremely common: see, e.g., Greene (2012) and Davidson and MacKinnon (2004) for traditional approaches under a more structural paradigm. Freedman (2009) provides a skeptical treatment of a structural approach to regression. Here, we discuss the articles and books that most closely relate to the agnostic paradigm. The readings from Chapter 2 remain highly relevant, particularly Hansen (2013), Goldberger (1991), and Angrist and Pischke (2009), all of which link closely to the presentation of materials here. Buja et al. (mimeo.) provides an in-depth article-length treatment of the properties of regression and associated standard errors when the CEF is not linear in the explanatory variables. Chen (2007) provides an elegant derivation of the asymptotic properties of a broad family of sieve estimators. Our book only considers the asymptotic properties of standard error estimators; Imbens and Kolesar (in press) provides practical recommendations for estimating standard errors with small samples. Cameron and Miller (2015) is a similarly practical article, considering robust standard error estimation under clustering. The idea of kernel estimation (Definition 3.3.7) can be wedded to regression to yield kernel regression (for example, the canonical Nadaraya–Watson estimator) and generalizations (such as LOESS). These can be thought of as locally weighted regression fits of observations near a given set of values for the explanatory variables, with larger weights assigned to observations closer to the point of prediction; 2006 provides a clear treatment. We strongly recommend James et al. (2014) and its more technical analog Hastie, Tibshirani, and Friedman (2009) for in-depth, book-length treatments of statistical prediction, including (penalized) regression and its generalizations. For article-length overviews of prediction methods with high-dimensional data, we recommend Varian (2014) and Belloni, Chernozhukov, and Hansen (2014).